Islam Asous

+1(917) 837-8908 | islam.asous1@gmail.com | islamasous.dev | linkedin.com/in/islam-asous | github.com/I-Asous

Education

Hunter College, City University of New York (CUNY) — New York, NY
Bachelor of Arts in Computer Science, Minor in Mathematics — *Expected Dec 2026*

- **Relevant Coursework:** Data Structures & Algorithms, Programming Methods, Computer Theory, Computer Architecture, Database Management, Foundations & Practice of Modern AI Systems, Automation Engineering, Linear Algebra

Experience

NYC Department of Transportation — Aug 2026 – Present
Software Engineer Intern — *Remote*

- Scoping and delivering a minimum viable wind load capacity map against a formal project scope and change-order process, translating stakeholder requirements into defined GIS deliverables and timelines.
- Digitizing NYC Building Code §1609.4.3 shoreline wind exposure figures across all five boroughs and classifying building footprints into Exposure B, C, and D using code distance thresholds (600–5,000 ft).
- Communicating progress through written status reports and oral presentations to DOT leadership, practicing stakeholder management and technical communication.

Project Sisyphus, Clemson University — June 2026 – Present
Software Engineer REU — *Remote*

- Maintain a authentication-tracking browser extension (***WXT, Vue 3, TypeScript, Dexie.js***) for NSF-funded research ($650K, Award #2521037) on authentication-fatigue-driven password reuse across enterprise platforms.
- Built a real-time broadcast notification system (***Vue 3, Pinia, Supabase Realtime***) enabling admins to push instant announcements, enforcing role-based access via Postgres Row-Level Security.
- Embedded ***Claude Code*** into an agentic development workflow spanning implementation, testing, and cross-repository debugging (Supabase, OAuth, CI), accelerating feature delivery while maintaining 90%+ test coverage.

CUNY Tech Prep — May 2026 – Present
Data Science Fellow — *New York, NY*

- Selected for a competitive fellowship with students from across the 11 CUNY senior colleges where Fellows create end-to-end data science and AI projects using tools such as Python, SQL, Pandas, scikit-learn, and Hugging Face.
- Participate in weekly courses and learn industry best practices for exploratory data analysis (EDA), feature engineering, statistical analysis, data visualization, LLMs, RAGs, vector databases, and generative AI applications.

Hunter College Computer Science Department — Aug 2025 – Present
Undergraduate Teaching Assistant — *New York, NY*

- Supported 600+ students across Computer Architecture I & II, Computer Theory, and OOP/Data Structures & Algorithms by leading weekly review sessions and office hours.
- Co-designed quizzes, midterms, and finals for courses of 150–300+ students; reduced grading turnaround by building automated evaluation tools with faculty.

Projects

Gridlock, Multi-Agent Specification-Gaming Testbed — Oct 2026 – Present
AI/ML Engineer — Foundations & Practice of Modern AI Systems(Course) — *Remote*

- Building a reproducible multi-agent gridworld that measures **specification gaming**: agents satisfying stated rules (R) while violating hidden designer intent (R*): extending the AI Safety Gridworlds framework with adversarial multi-agent scenarios.
- Train a **tabular Q-learning agent** that discovers rule-legal "loopholes" through trial & error, benchmarked against **A*** path-planning and **Constraint-Satisfaction** agents behind a shared interface to isolate how agent type affects gaming.
- Engineer an **oversight monitor** that scores behavior against hidden intent (R*) and flags divergence, evaluated by detection rate, time-to-detect, and false-positive rate against a non-ML static-rule baseline and a trace-aware vs. action-only ablation.

Token Watching, CUNY Hunter — Aug 2026 – Present
AI/Backend Engineer — Capstone — *Remote*

- Leading the agent workstream on a 4-person team, building a multi-agent pipeline (Auditor → Optimizer → Validator) served via **FastAPI** to a Chrome extension that cuts token waste in ChatGPT and Claude prompts.
- Cut pipeline cost **48**× and p95 latency **7**× (36.6s → 5.3s) on a 1,000-prompt eval set by running local NLP checks (***spaCy, textstat, Tiktoken***) before any model call and using **Claude** tool use with strict JSON schemas and prompt caching.
- Raised the Validator's first-attempt pass rate from 67% to **98.6%**, as scored by a 1–10 grounded LLM judge, by tracing every Claude call, tool call, and verdict in **LangSmith** to debug failures across 6 eval iterations.

Technical Skills

Programming: C++, Python, HTML/CSS, TypeScript, JavaScript
Frameworks, Cloud & Tools: Git/GitHub, React, Node.js, Express.js, Supabase, Docker, REST APIs
Certifications: AWS Skill Builder - Developer, Cloud Operator (SysOps), DevOps (In Progress); CodePath AI201: Applications of AI Engineering (In Progress);